

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Mr. Rawson • Computer Science



Year 10 2026

# Pixel Art and Colour Depth

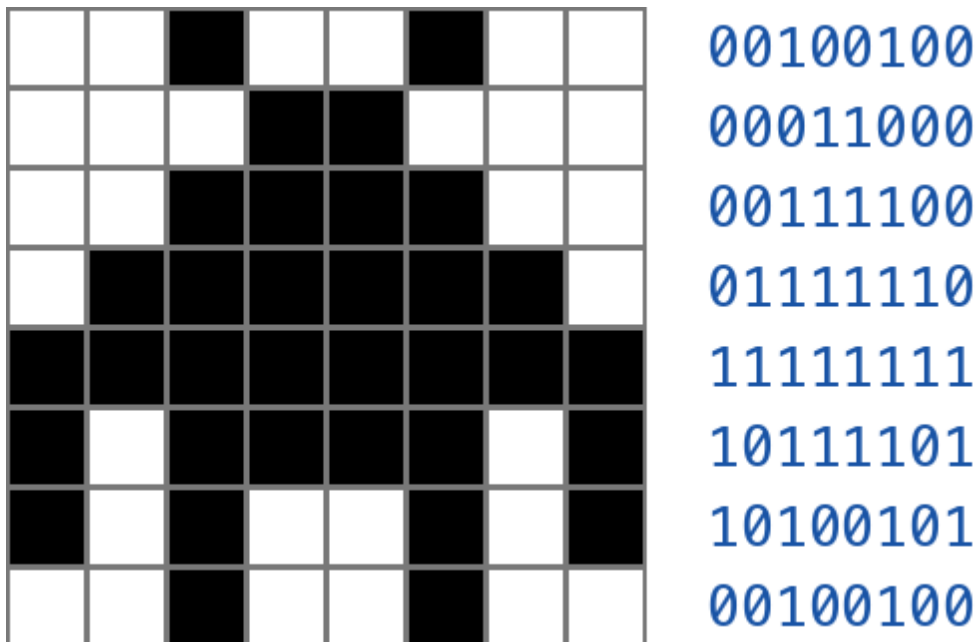
## LO1 • A picture is a grid of pixels, and each pixel is a number

*OCR 1.2.4 — representing images · 1-bit colour depth*

Below is a picture 8 pixels wide and 8 pixels tall. Two things turn it into binary, and both of them are always given to you.

**The index**, which says what each pattern means. Here 0 is white and 1 is black.

**The order**, which starts at the **top left** and runs **left to right, top to bottom**, the way you read a page.



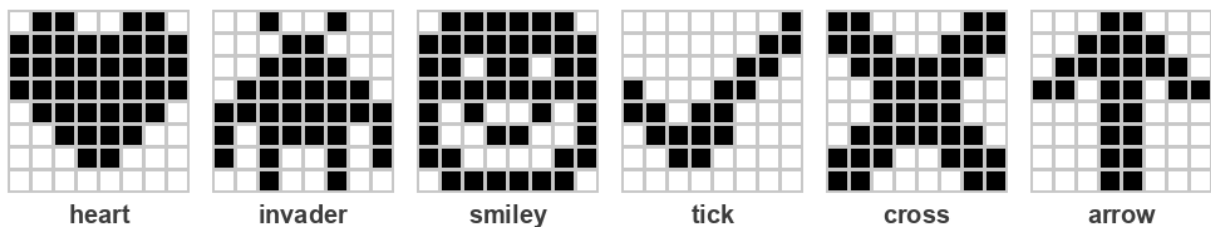
Two colours need only one bit per pixel, so this is a **colour depth of 1**. Every row is 8 bits, and there are 8 rows.

$8 \times 8 = 64$  pixels, 1 bit each, so 64 bits, which is  $64 \div 8 = 8$  bytes.

## LO2 • The same idea, six more times

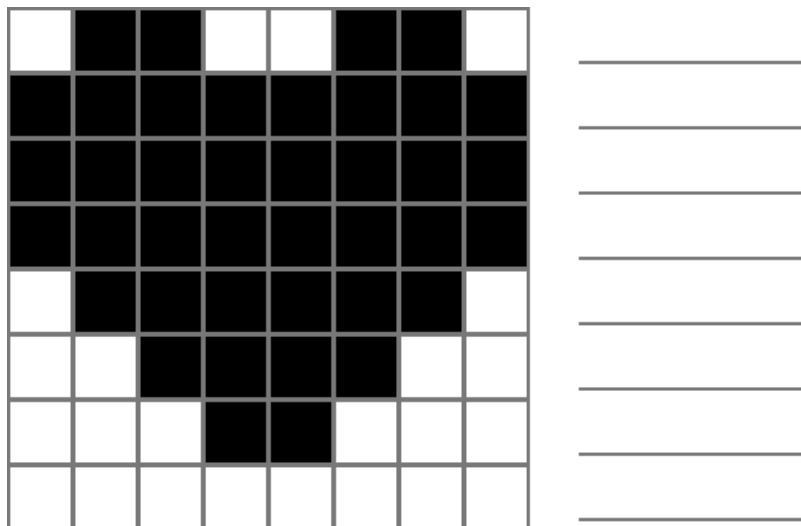
*OCR 1.2.4 — encoding and decoding a bitmap*

Every one of these is 8 by 8, at a colour depth of 1.



- (a) Write the binary for each row of this picture. The index is the same as before, 0 white and 1 black.

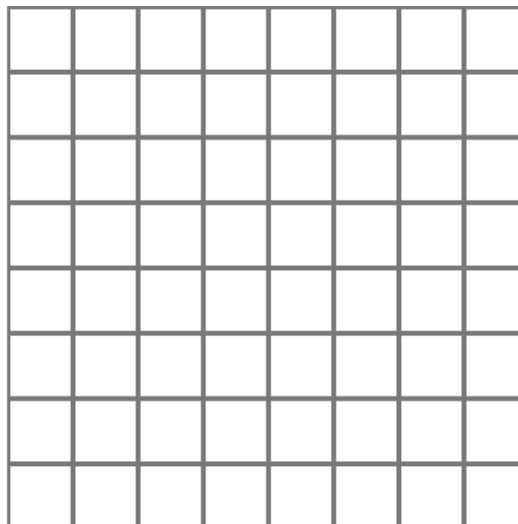
[4]



- (b) Now the other way round. Shade the grid to read this picture back out.

[4]

00000001  
00000011  
00000110  
10001100  
11011000  
01111000  
00110000  
00000000



*Check:* the shaded squares make a picture you recognise.

- (c) Both pictures above are 8 pixels by 8 pixels at a colour depth of 1. State how many bytes each one needs.

[2]

---

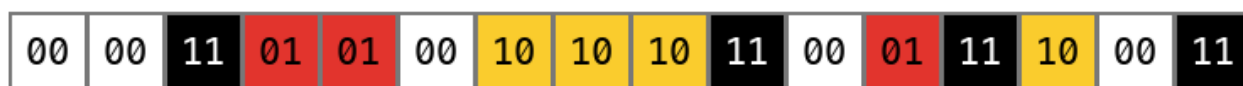
### LO3 • More than two colours, so the index does more work

---

#### OCR 1.2.4 — indexed bitmaps · 2-bit colour depth

With four colours, one bit per pixel is no longer enough. Four colours need **2 bits** each, because  $2^2 = 4$ , and the **index** says which pattern means which colour. Here it is 00 white, 01 red, 10 yellow, 11 black.

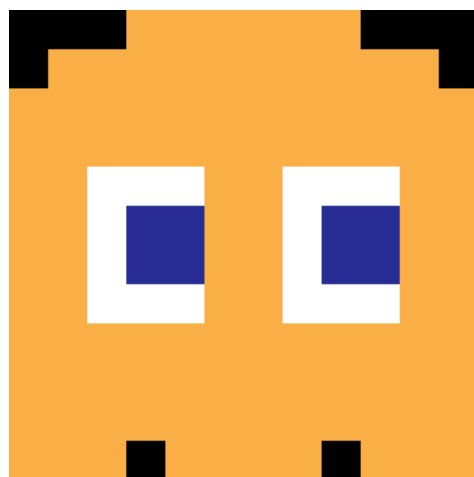
Writing each pixel's code onto it turns the picture straight into the bits.



00001101010010101011000111100011

Sixteen pixels, 2 bits each, so 32 bits, which is  $32 \div 8 = 4$  bytes.

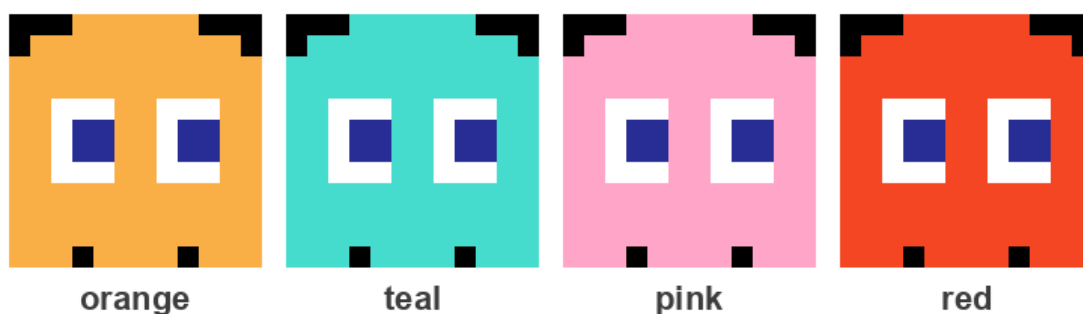
The same thing works on a whole picture rather than one row. This ghost uses **exactly four colours** — background, body, eye white and pupil — so it too is a **colour depth of 2**.



#### index

|    |                                                                                      |            |
|----|--------------------------------------------------------------------------------------|------------|
| 00 |  | background |
| 01 |  | body       |
| 10 |  | eye white  |
| 11 |  | pupil      |

Here is the part worth remembering. All four of these ghosts are stored with **exactly the same bits**. Only the index changed.



So the bits on their own do not tell you what the picture looks like. That is why an exam question always prints the index next to the data, and why reading it off is the first thing to do.

- (d) The ghost is 12 pixels wide and 12 pixels tall, at a colour depth of 2. Calculate its file size in bytes.

[3]

- (e) A designer wants to add a fifth colour to the ghost. State the colour depth this now needs, and give the number of colours that depth makes available.

[2]

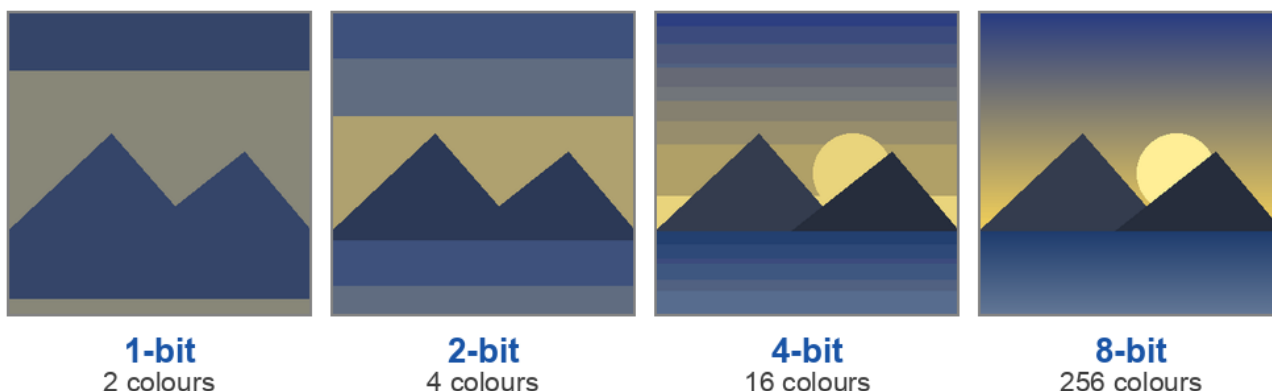
## LO4 • More bits buy more colours

### OCR 1.2.4 — colour depth and its effect on quality

Colour depth is the number of bits stored for one pixel. Those bits work like the place-value columns from Booklet 1, so  $n$  bits give  $2^n$  colours. It is the same doubling table you already have.

| Colour depth      | 1 bit     | 2 bits    | 3 bits    | 4 bits     | 8 bits      |
|-------------------|-----------|-----------|-----------|------------|-------------|
| Colours available | $2^1 = 2$ | $2^2 = 4$ | $2^3 = 8$ | $2^4 = 16$ | $2^8 = 256$ |

Here is one picture stored four times, at four different colour depths.



And here is the formula, worth writing down in exactly this form because it is short enough to remember under pressure.

$$\text{Size (B)} = \text{W} \times \text{H} \times \text{D} \div 8$$

- (f) Look along the four pictures above. Describe what more colour depth gives you, and say what it costs.

[3]

- (g) The 1-bit picture lost its sun completely, and the flat-coloured ghost on the previous page looks perfect at only 2 bits. Explain why the two pictures cope so differently with a low colour depth.

[3]

- (h) An image is 200 pixels wide, 150 pixels tall, and has a colour depth of 8 bits. Calculate its file size in bytes.

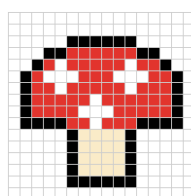
[3]

## LO5 • Your turn

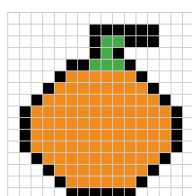
*Piskel · piskelapp.com · free, in the browser, no account needed*

Open [piskelapp.com](https://piskelapp.com) and choose **Create Sprite**. Set the canvas to **32 × 32**. (If it already opens at 32 by 32, you are ready to draw.)

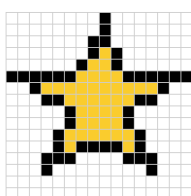
Pick one subject and build it twice, once at each colour budget below. Keep the same subject both times, because the comparison is between the two budgets rather than between two different pictures. The third budget is shown to you at the end.



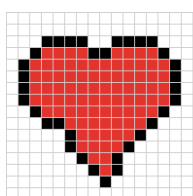
mushroom



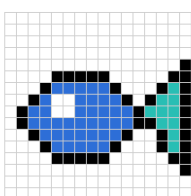
pumpkin



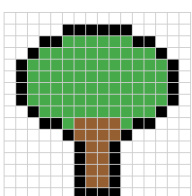
star



heart



fish



tree

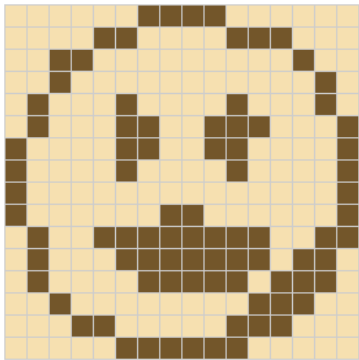
- (i) **Build 1 — 1-bit.** Use **exactly 2 colours**.

*Check:* count the swatches you have used. Two means a colour depth of 1.

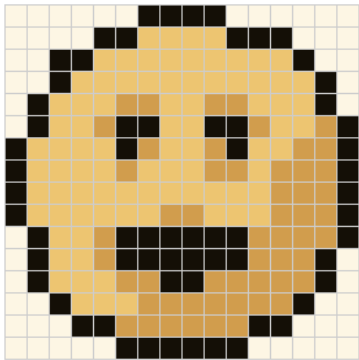
(j) **Build 2 — 2-bit.** Build it again with **exactly 4 colours**, then write your index here, the way the ghost’s index was written.

|        |    |    |    |    |
|--------|----|----|----|----|
| Code   | 00 | 01 | 10 | 11 |
| Colour |    |    |    |    |

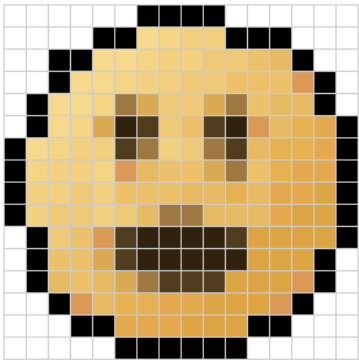
(k) **Build 3 is done for you.** Sixteen colours takes a long while to choose and longer still to fill in, so here is the same picture already built at all three budgets.



1-bit  
2 colours



2-bit  
4 colours



4-bit  
16 colours

Look along the three. Describe one thing the 16-colour version shows that the 4-colour version cannot.

[2]

(l) Your two builds and the example above are all 32 by 32, so each one holds  $32 \times 32 = 1024$  pixels. Complete the table.

[3]

| Picture     | Colour depth | Total bits | Size in bytes |
|-------------|--------------|------------|---------------|
| Build 1     |              |            |               |
| Build 2     |              |            |               |
| The example |              |            |               |

(m) Look down the final column. Describe what happens to the file size each time one bit is added to the colour depth.

[2]